

Extended Curriculum Vitae

for *Rúben Alexandre Dinis Araújo*

(Ph.D. Biomedicine, NOVA Medical School)

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Disclaimer

- 1) Additional evidence, publications, awards and supporting documentation are available at: <https://rubenaraujo.org/cv/>. The website also hosts additional information such as academic degrees, diplomas, professional experience and other relevant personal and scientific background.
- 2) The majority of the research presented in this document was carried out across three locations: the *Laboratório de Engenharia e Saúde* at *Instituto Superior de Engenharia de Lisboa* (ISEL); the Intensive Care Unit of *Centro Hospitalar Universitário de Lisboa Central* (CHULC); and the Laboratory of Serology at *Instituto Português do Sangue e da Transplantação* (IPST), located within *Pulido Valente* Hospital.

Summary

This detailed *Curriculum Vitae* refers (mostly) to work developed from 2020 onwards, starting from the time the candidate applied to the Master in Biomedical Engineering in *Instituto Superior de Engenharia de Lisboa* (ISEL), and eventually the research conducted within the Doctoral Programme in Biomedicine at NOVA Medical School (NMS) | *Faculdade de Ciências Médicas* (FCM), in association with *Universidade NOVA de Lisboa* and *Universidade de Aveiro*, with its recent conclusion in April 20, 2026.

The candidate's research mainly focuses on biomarker-based diagnostics and prognostics in critically ill patients, integrating metabolomics through high-throughput serum profiling using FTIR spectroscopy and targeted proteomics through multiplex cytokine analysis, combined with advanced statistical and machine-learning models. It also highlights the identification of molecular and immune signatures associated with infection status, clinical deterioration and mortality risk in the Intensive Care Unit (ICU), and the development of translational tools that can support earlier and more precise clinical decision-making in acute care and potentially in future respiratory pandemics.

The candidate's other works ranges from human embryo viability (in collaboration with Mallot Clinic), genotoxicity studies (in collaboration with former *Escola Superior de Tecnologia da Saúde de Lisboa – ESTeSL* – now *Escola Superior de Saúde de Lisboa*), proteomic studies with UHPL-HRMS data (in collaboration with IST and University of Ferrara – UNIFE – in Italy). A big focus of work also lies in the field of organ transplantation and rejection, in collaboration with *Instituto Português do Sangue e da Transplantação – IPST*.

Following the **Disclaimer** and this **Summary**, **Chapter 1** presents the first-author publications. **Chapter 2** compiles co-authored publications developed within the PREMO project ("Predictive Models of COVID-19 Outcomes for Higher Risk Patients Towards a Precision Medicine"; FCT reference **DSAIPA/DS/0117/2020**), while **Chapter 3** lists other co-authored publications produced in collaboration with either established researchers or other (still) Ph.D. candidates/finalists either from the same doctoral programme or from related biomedical areas. **Chapter 4** summarizes conference outputs, structured into three subchapters—**Papers & Abstracts**, **Posters** and **Oral Communications by Invitation**—reporting titles, conference names, dates and, when available, DOIs or conference URLs. **Chapter 5** summarizes all Financed Projects participation (in table form). **Chapter 6** pertains to relevant experience in Innovation and Technology Transfer. **Chapter 7** briefly details various subjects from supervision and mentoring activities to scientific outreach efforts. The final chapter, **Awards & Distinctions**, provides a concise synthesis of the entire document.

Research Keywords: Intensive Care Unit; COVID-19; Biomarkers; FTIRS; Cytokines; Machine Learning; Infection Diagnostics; Mortality Prediction; Clinical Decision Support; Precision Medicine.

Chapter 1 – First-author Publications

The following list includes all first-author publications, with the first 8 directly related to the doctoral thesis. The works below are listed from most recent to oldest. All entries are published with the exception of **Manuscript 1**.

Manuscript 1

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Unlocking infection biomarkers through serum molecular fingerprint in critically ill patients. **Manuscript withheld due to ongoing provisional patent application (P2134.1 PP)**

Manuscript 2

Araújo, R.; Von Rekowski, C.P.; Fonseca, T.A.H.; Calado, C.R.C.; Ramalhete, L.; Bento, L. Multiplex Targeted Proteomic Analysis of Cytokine Ratios for ICU Mortality in Severe COVID-19. *Proteomes* 2025, 13, 35. <https://doi.org/10.3390/proteomes13030035>

Manuscript 3

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Calado, C.R.C.; Bento, L. Cytokine-Based Insights into Bloodstream Infections and Bacterial Gram Typing in ICU COVID-19 Patients. *Metabolites* 2025, 15, 204. <https://doi.org/10.3390/metabo15030204>

Manuscript 4

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Early Mortality Prediction in Intensive Care Unit Patients Based on Serum Metabolomic Fingerprint. *Int. J. Mol. Sci.* 2024, 25, 13609. <https://doi.org/10.3390/ijms252413609>

Manuscript 5

Araújo, R.; Viegas, A.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Discovery of Delirium Biomarkers through Minimally Invasive Serum Molecular Fingerprinting. *Metabolites* 2024, 14, 301. <https://doi.org/10.3390/metabo14060301>

Manuscript 6

Araújo, R.; Ramalhete, L.; Viegas, A.; Von Rekowski, C.P.; Fonseca, T.A.H.; Calado, C.R.C.; Bento, L. Simplifying Data Analysis in Biomedical Research: An Automated, User-Friendly Tool. *Methods Protoc.* 2024, 7, 36. <https://doi.org/10.3390/mps7030036>

Manuscript 7

Araújo, R.; Ramalhete, L.; Ribeiro, E.; Calado, C.R.C. Plasma versus Serum Analysis by FTIR Spectroscopy to Capture the Human Physiological State. *BioTech* **2022**, *11*, 56. <https://doi.org/10.3390/biotech11040056>

Manuscript 8

Araújo, R.; Bento, L.; Fonseca, T.A.H.; Von Rekowski, C.P.; da Cunha, B.R.; Calado, C.R.C. Infection Biomarkers Based on Metabolomics. *Metabolites* **2022**, *12*, 92. <https://doi.org/10.3390/metabo12020092>

Manuscript 9

Araújo, R.; Ramalhete, L.; da Paz, H.; Ladeira, C.; Calado, C.R.C. A New Method to Predict Genotoxic Effects Based on Serum Molecular Profile. *Spectrochim. Acta A Mol. Biomol. Spectrosc.* **2021**, *255*, 119680. <https://doi.org/10.1016/j.saa.2021.119680>

Manuscript 10

Araújo, R.; Ramalhete, L.; Da Paz, H.; Ribeiro, E.; Calado, C.R.C. A Simple, Label-Free, and High-Throughput Method to Evaluate the Epigallocatechin-3-Gallate Impact in Plasma Molecular Profile. *High-Throughput* **2020**, *9*, 9. <https://doi.org/10.3390/ht9020009>

Chapter 2 – Co-authored Publications (PREMO Project)

This chapter compiles the candidate's co-authored contributions conducted within the PREMO project ("Predictive Models of COVID-19 Outcomes for Higher Risk Patients Towards a Precision Medicine"; FCT reference DSAIPA/DS/0117/2020). The PREMO initiative, developed across CHULC, ISEL and *Instituto Superior Técnico – Investigação, Desenvolvimento e Inovação em Ciência e Tecnologia* (IST-ID), focused on constructing robust statistical and machine-learning models to classify and predict severe COVID-19 outcomes, integrating clinical, therapeutic, laboratory, and longitudinal survival data.

More information about the project is available at: <https://www.isel.pt/investigacao-e-inovacao/ecossistema-de-inovacao/projetos/2020/premo>. The works listed here reflect collaborative efforts with other researchers in the Biomedicine Doctoral Programme and associated laboratory teams.

Manuscript 11

Von Rekowski, C.P.; Fonseca, T.A.H.; Araújo, R.; Calado, C.R.C.; Bento, L.; Pinto, I. Blood biomarker trajectories in ICU-directed prediction models – A scoping review. *EPMA Journal* **17**, 427–455 (2026). <https://doi.org/10.1007/s13167-026-00456-5>

Manuscript 12

Viegas, A.; Von Rekowski, C.P.; Araújo, R.; Viana-Baptista, M.; Macedo, M.P.; Bento, L. Predicting ICU Delirium in Critically Ill COVID-19 Patients Using Demographic, Clinical, and Laboratory Admission Data: A Machine Learning Approach. *Life* 2025, 15, 1045. <https://doi.org/10.3390/life15071045>

Manuscript 13

Von Rekowski, C.P.; Fonseca, T.A.H.; Araújo, R.; Martins, A.; Pinto, I.; Oliveira, M.C.; Justino, G.C.; Bento, L.; Calado, C.R.C. Predictive Models of Patient Severity in Intensive Care Units Based on Serum Cytokine Profiles: Advancing Rapid Analysis. *Appl. Sci.* 2025, 15, 4823. <https://doi.org/10.3390/app15094823>

Manuscript 14

Fonseca, T.A.H.; Von Rekowski, C.P.; Araújo, R.; Oliveira, M.C.; Justino, G.C.; Bento, L.; Calado, C.R.C. Comparison of two metabolomics-platforms to discover biomarkers in critically ill patients from serum analysis. *Computers in Biology and Medicine* 2025;184:109393. <http://dx.doi.org/10.1016/j.compbiomed.2024.109393>

Manuscript 15

Von Rekowski, C.P.; Pinto, I.; Fonseca, T.A.H.; Araújo, R.; Calado, C.R.C.; Bento, L. Analysis of six consecutive waves of ICU-admitted COVID-19 patients: key findings and insights from a Portuguese population. *GeroScience* 47, 2399–2422 (2025). <https://doi.org/10.1007/s11357-024-01410-x>

Manuscript 16

Caldeira, V.; Fonseca, T.A.H.; N'Dembo, Luana; Araújo, R.; Von Rekowski, C.P.; Sampaio, P.N.S.; Calado, C.R.C. A new methodology for a rapid and high-throughput comparison of molecular profiles and biological activity of phytoextracts. *Biotech & Bioengineering* 2024;121(10):3047–58. <http://dx.doi.org/10.1002/bit.28739>

Manuscript 17

Von Rekowski, C.P.; Fonseca, T.A.H.; Araújo, R.; Brás-Geraldes, C.; Calado, C.R.C.; Bento, L.; Pinto, I. The Characteristics and Laboratory Findings of SARS-CoV-2 Infected Patients during the First Three COVID-19 Waves in Portugal—A Retrospective Single-Center Study. *Medicina* 2024, 60, 59. <https://doi.org/10.3390/medicina60010059>

Manuscript 18

Ladeira, C.; Araújo, R.; Ramalhete, L.; Teixeira, H.; Calado, C.R.C. Blood molecular profile to predict genotoxicity from exposure to antineoplastic drugs. *Mutation Research/Genetic Toxicology and Environmental Mutagenesis* 2023;891:503681. <http://dx.doi.org/10.1016/j.mrgentox.2023.503681>

Manuscript 19

Fonseca, T.A.H.; Von Rekowski, C.P.; Araújo, R.; Oliveira, M.C.; Justino, G.C.; Bento, L.; Calado, C.R.C. The Impact of the Serum Extraction Protocol on Metabolomic Profiling Using UPLC-MS/MS and FTIR Spectroscopy. *ACS Omega* 2023;8(23):20755–66. <http://dx.doi.org/10.1021/acsomega.3c01370>

Chapter 3 – Co-authored Publications (others)

This chapter includes co-authored publications unrelated to either to the Biomedicine doctoral programme or the PREMO project. These works were developed in collaboration with other Ph.D. candidates/finalists and researchers in their respective fields, ranging from Delirium in ICU COVID-19 patients at CHULC, to immunology and transplantation research at IPST.

Manuscript 20

Ramalhete, L.; Araújo, R.; Vigia, E.; Vieira, M.B.; Ferreira, A.; Calado, C.R.C. Risk Phenotyping Before Graft Implantation: FTIR Spectroscopy and Machine Learning for Complementary Risk Stratification in Kidney Transplantation. *Med. Sci.* **2026**, 14, 353. <https://doi.org/10.3390/medsci14030353>

Manuscript 21

Vigia, E.; Ramalhete, L.; Araújo, R.; Corado, S.; Barros, I.; Chumbinho, B.; Nobre, A.; Carrelha, S.; Pico, P.; Rodrigues, F.; et al. Day-Zero Serum FTIR Spectroscopy Identifies a Biochemical Signature Associated with Functional Pancreas Graft Dysfunction After Simultaneous Pancreas–Kidney Transplantation. *Life* **2026**, 16, 1054. <https://doi.org/10.3390/life16071054>

Manuscript 22

Ramalhete, L.; Oliveira, V.; Quintas, R.; Araújo, R. Exploratory Image-Level Classification of a Public Chest Radiograph Dataset Using a Lightweight SqueezeNet-Based Pipeline. *AI Med.* **2026**, 1, 15. <https://doi.org/10.3390/aimed1020015>

Manuscript 23

Vigia, E.; Ramalhete, L.; Araújo, R.; Corado, S.; Barros, I.; Chumbinho, B.; Nobre, A.; Carrelha, S.; Pico, P.; Rodrigues, F.; et al. Pre-Transplant Serum FTIRS Signatures as Predictive Biomarkers of Early Transient Pancreatic Graft Dysfunction in Simultaneous Pancreas–Kidney Transplantation. *Life* **2026**, 16, 780. <https://doi.org/10.3390/life16050780>

Manuscript 24

Ramalhete, L.; Araújo, R.; Vieira, M.B.; Vigia, E.; Calado, C.R.C.; Ferreira, A. Rejection-Focused Precision Medicine in Kidney Transplantation: Biology, Biomarkers, and Artificial Intelligence. *Life* **2026**, *16*, 674. <https://doi.org/10.3390/life16040674>

Manuscript 25

Ramalhete, L.; Araújo, R.; Vieira, M.B.; Vigia, E.; Pena, A.; Carrelha, S.; Teixeira, C.; Ferreira, A.; Calado, C.R.C. Machine Learning-Assisted FTIR Spectroscopy Analysis of Kidney Preservation Fluids for Delayed Graft Function Risk Stratification. *J. Clin. Med.* **2026**, *15*, 2762. <https://doi.org/10.3390/jcm15072762>

Manuscript 26

Ramalhete, L.; Almeida, P.; Araújo, R.; Espada, E. Integrative Population Analysis of MICA and MICB Using Unsupervised Machine Learning in a Large Histocompatibility Laboratory Cohort. *J* **2026**, *9*, 8. <https://doi.org/10.3390/j9010008>

Manuscript 27

Ramalhete, L.; Araújo, R.; Teixeira, C.; Pedro, I.; Silva, I.; Ferreira, A. Mapping Anti-HLA Class I Cross-Reactivity for Transplantation Using Interpretable Embedding and Clustering of SAB MFI. *AI Med.* **2025**, *1*, 1. <https://doi.org/10.3390/aimed1010001>

Manuscript 28

Ramalhete, L.; Araújo, R.; Vieira, M.B.; Vigia, E.; Pena, A.; Carrelha, S.; Ferreira, A.; Calado, C.R.C. Rapid FTIR Spectral Fingerprinting of Kidney Allograft Perfusion Fluids Distinguishes DCD from DBD Donors: A Pilot Machine Learning Study. *Metabolites* **2025**, *15*, 702. <https://doi.org/10.3390/metabo15110702>

Manuscript 29

Viegas, A.; Von Rekowski, C.P.; Araújo, R.; Ramalhete, L.; Cordeiro, I.M.; Manita, M.; Viana-Baptista, M.; Macedo, P.; Bento, L. Predicting Delirium in Critically Ill COVID-19 Patients Using EEG-Derived Data: a Machine Learning Approach. *GeroScience* (2025). <https://doi.org/10.1007/s11357-025-01809-0>

Manuscript 30

Ramalhete, L.; Araújo, R.; Vieira, M.B.; Vigia, E.; Aires, I.; Ferreira, A.; Calado, C.R.C. Integration of FTIR Spectroscopy and Machine Learning for Kidney Allograft Rejection: A Complementary Diagnostic Tool. *J. Clin. Med.* **2025**, *14*, 846. <https://doi.org/10.3390/jcm14030846>

Manuscript 31

Ramalhete, L.; Vieira, M.B.; Araújo, R.; Vigia, E.; Aires, I.; Ferreira, A.; Calado, C.R.C. Predicting Cellular Rejection of Renal Allograft Based on the Serum Proteomic Fingerprint. *Int. J. Mol. Sci.* **2024**, *25*, 3844. <https://doi.org/10.3390/ijms25073844>

Manuscript 32

Ramalhete, L.; Almeida, P.; Ferreira, R.; Abade, O.; Teixeira, C.; Araújo, R. Revolutionizing Kidney Transplantation: Connecting Machine Learning and Artificial Intelligence with Next-Generation Healthcare—From Algorithms to Allografts. *BioMedInformatics* 2024, 4, 673-689. <https://doi.org/10.3390/biomedinformatics4010037>

Manuscript 33

Ramalhete, L.; Araújo, R.; Teixeira, C.; Almeida, P.; Silva, I.; Lima, A. Evaluation of rapid optimized flow cytometry crossmatch (Halifaster) in living donor kidney transplantation. *HLA* 2024;103(2):e15391. <http://dx.doi.org/10.1111/tan.15391>

Manuscript 34

Ramalhete, L.; Araújo, R.; Ferreira, A.; Calado, C.R.C. Exosomes and microvesicles in kidney transplantation: the long road from trash to gold. *Pathology* 2024;56(1):1–10. <http://dx.doi.org/10.1016/j.pathol.2023.10.004>

Manuscript 35

Ramalhete, L.; Vigia, E.; Araújo, R.; Marques, H.P. Proteomics-Driven Biomarkers in Pancreatic Cancer. *Proteomes* 2023, 11, 24. <https://doi.org/10.3390/proteomes11030024>

Manuscript 36

Ramalhete, L.; Araújo, R.; Ferreira, A.; Calado, C.R.C. Label-free discrimination of T and B lymphocyte activation based on vibrational spectroscopy – A machine learning approach. *Vibrational Spectroscopy* 2023;126:103529. <http://dx.doi.org/10.1016/j.vibspec.2023.103529>

Manuscript 37

Ramalhete, L.M.; Araújo, R.; Ferreira, A.; Calado, C.R.C. Proteomics for Biomarker Discovery for Diagnosis and Prognosis of Kidney Transplantation Rejection. *Proteomes* 2022, 10, 24. <https://doi.org/10.3390/proteomes10030024>

Chapter 4 – Conference Publications

This chapter summarises all conference outputs. It is organised into three subchapters—conference **Papers & Abstracts**, **Posters**, and **Oral Communications by Invitation**, each listing the title of the contribution, conference name, and date. When available, DOIs or conference URLs are provided. Only one of the invited oral communications was directly linked to the doctoral research; the remaining invitations were related to broader academic activities, including serving as Session Moderator at the **XIV Meeting of Quality Researchers**

(RIQUAL, ISEL) and co-presenting work at **NOVA Summer Medical School**. As stated in the **Disclaimer**, all conference materials may also be consulted in the candidate's website. All first-author conference contributions listed in this chapter (underlined) correspond exclusively to work directly related to the PhD in Biomedicine, while underlined non-first-author contributions pertain to the PREMO project. Remainder communications, like posters, relate to work done during the Masters in Biomedical Engineering.

Papers & Abstracts

Conference Paper 1

Araújo, R.; Ramalhete, L.; Fonseca, T.A.H.; Von Rekowski, C.P.; Bento, L.; Calado, C.R.C. Infection Biomarkers at Intensive Care Units. 4th ROME International Conference on Challenges in Engineering, Medical, Economics and Education: Research & Solutions (CEMEERS24b), Rome, 2024

Conference Paper 2

Correia, I; Araújo, R.; Fonseca, T.A.H.; Von Rekowski, C.P.; Bento, L.; Calado, C.R.C. Survival Prediction in Critically Ill Patients based on the Serum Molecular Fingerprint. 28th LISBON International Conference on Science, Engineering, Technology and Healthcare (SETH-24) Dec. 16-18, 2024 Lisbon (Portugal)

Conference Paper 3

Correia, I; Fonseca, T.A.H.; Pataco, J.; Oliveira, M.C.; Caldeira, V.; Domingues, N.; Von Rekowski, C.P.; Araújo, R.; Bento, L.; Calado, C.R.C. Comparison of the Serum Whole Molecular Composition with the Serum Metabolome to Acquire the Pathophysiological State. 4th ROME International Conference on Challenges in Engineering, Medical, Economics and Education: Research & Solutions (CEMEERS24b), Rome, 2024

Conference Paper 4

Von Rekowski, C.P.; Pinto, I.; Fonseca, T.A.H.; Araújo, R.; Ferreira, A.; Calado, C.R.C.; Bento, L. Trends in COVID-19 Patient Characteristics and Mortality Throughout the Pandemic: Insights from a Portuguese Single Centre Study. 4th ROME International Conference on Challenges in Engineering, Medical, Economics and Education: Research & Solutions (CEMEERS24b), Rome, 2024

Conference Paper 5

Fonseca, T.A.H.; Von Rekowski, C.P.; Araújo, R.; Oliveira, M.C.; Bento, L.; Justino, G.C.; Calado, C.R.C. Predicting Critically Ill Patients Outcome in the ICU using UHPLC-HRMS Data. 3rd International Conference on Challenges in Engineering, Medical, Economics and Education: Research & Solutions (CEMEERS-24a), Porto, 2024

Conference Paper 6

Justino, G.C.; Fonseca, T.A.H.; Von Rekowski, C.P.; Araújo, R.; Oliveira, M.C.; Bento, L.; Calado, C.R.C. Enhancing the preparedness toolbox through a post-omics expedition - unveiling the COVID molecular signature on biochemical pathways. XII National Congress of Biochemistry, Aveiro, 2024

Conference Paper 7

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Alternative serum biomarkers of bacteraemia for intensive care unit patients. 2023 IEEE 7th Portuguese Meeting on Bioengineering, ENBENG 2023. Institute of Electrical and Electronics Engineers (IEEE), 22-23 June 2023, Porto University, Porto, Portugal.

Conference Paper 8

Von Rekowski, C.P.; Araújo, R.; Fonseca, T.A.H.; Calado, C.R.C.; Bento, L.; Pinto, I. Laboratory biomarkers associated to death in the first three COVID-19 waves in Portugal. 2023 IEEE 7th Portuguese Meeting on Bioengineering, ENBENG 2023. Institute of Electrical and Electronics Engineers (IEEE), 22-23 June 2023, Porto University, Porto, Portugal.

Conference Paper 9

Fonseca, T.A.H.; Oliveira, M.C.; Araújo, R.; Bento, L.; Von Rekowski, Justino, G.C. Comparison of Analytical Methods Of Serum Untargeted Metabolomics. 2023 IEEE 7th Portuguese Meeting on Bioengineering, ENBENG 2023. Institute of Electrical and Electronics Engineers (IEEE), 22-23 June 2023, Porto University, Porto, Portugal.

Abstract 1

Ramalhete, L.; Teixeira, A.; Teixeira, C.; Silva, I.; Almeida, P.; Lima, A.; Araújo, R. Immunogenetic profiling in living donor kidney transplantation: Insights from DSAs and pronase-treated flow cytometry crossmatch. 37th European Immunogenetics & Histocompatibility Conference (EFI) May 20-23, 2024 Geneva (Switzerland).

Abstract 2

Ramalhete, L.; Teixeira, C.; Silva, I.; Araújo, R. Deciphering HLA antibody reactivity patterns: A cluster-based analysis of SAB assay data. 37th European Immunogenetics & Histocompatibility Conference (EFI) May 20-23, 2024 Geneva (Switzerland).

Posters

Poster 1

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Predicting Septic Patient Outcomes Based on Serum Cytokines. 10th Edition Forum of Chemical & Biological Engineering. Instituto Superior de Engenharia de Lisboa (ISEL), 7-10 May 2024, Lisbon, Portugal.

Poster 2

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Streamline bacterial infection diagnosis: rapid Gram classification using FTIR spectroscopy. ChemPor 2023 - The International

Chemical and Biological Engineering Conference. 12-15 September, Bragança, Portugal. AMAQ Silva, AMCL Peres, AME Ribeiro, MOAS Ferreira, MFF Barreiro, PMP Brito, SPA Pinho, editors, Instituto Politécnico Bragança Editora, Bragança, ISBN 978-972-745-327-6.

Poster 3

Araújo, R.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Impact of COVID-19 on critically ill patients: mortality prediction models based on serum FTIR-spectra. 12th International Conference on Advanced Vibrational Spectroscopy (ICAVS12). 27th August - 1st September 2023, Krakow, Poland.

Poster 4

Fonseca, T.A.H.; Von Rekowski, C.P.; Araújo, R.; Bento, L.; Calado, C.R.C. Impact Of Serum Metabolome Isolation Process On Models' Prediction Of Critically Ill Patients' Mortality. 12th International Conference on Advanced Vibrational Spectroscopy (ICAVS12). 27th August - 1st September 2023, Krakow, Poland.

Poster 5

Araújo, R. Comparison of Outcomes Between the DEKA Arm and Conventional Prosthesis – an in-Depth Review Study of Measures and Evaluation Techniques. *IV Jornadas de Ortoprotesia da ESTEsL*. June 2, 2018, Lisbon, Portugal.

Oral Communications by Invitation

Oral Communication 1

Araújo, R.; Ramalhete, L. Big Data and Artificial Intelligence in Healthcare. NOVA Summer Medical School: Lifestyle, July 3, 2025 Lisbon (Portugal)

Oral Communication 2

Araújo, R. Session moderator on "Techniques and Optimization Methodologies". *XIV Encontro de Investigadores da Qualidade*, June 28, 2024 Lisbon (Portugal)

Oral Communication 3

Araújo, R.; Ramalhete, L.; Von Rekowski, C.P.; Fonseca, T.A.H.; Bento, L.; Calado, C.R.C. Fungal and Bacterial Infections discrimination in ICU patients based on serum molecular fingerprint. 4th Comprehensive Health Research Centre (CHRC) Annual Summit. 25-26 May 2023, University of Évora, Évora, Portugal.

Chapter 5 – Financed Projects

This chapter summarises, in table form, all projects the candidate has been involved in, while during the Master in Biomedical Engineering in ISEL, the Ph.D. in Biomedicine in NMS, as well as Principal Investigator (PI) in *Instituto Português da Face* (IPF). It does not include other projects that did not secure funding. The full list and URL links can be found at <https://rubenaraujo.org/cv/> under “Research” and “Projects”.

Project Title	Funding Type	Project ID	Role
Arthur: 3D Dentofacial Surgery Full Planning	Grant (479,839€)	POCI-01-0247-FEDER-039690	Researcher
Biodiscus	Grant (192,730€)	CENTRO-01-0247-FEDER-039969	Principal Researcher
GenTox	Grant (5,000€)	IDI&CA/IPL/2017/GenTox/ESTeSL	Researcher
Renal Prognosis	Grant (5,000€)	IDI&CA/IPL/2018/RenalProg/ISEL	Researcher
NephroMD	Grant (5,000€)	IDI&CA/IPL/2020/NephroMD:ISEL	Researcher
DrugsPlatf	Grant (5,000€)	IDI&CA/IPL/2017/DrugsPlatf/ISEL	Researcher
Predictive Models of COVID-19 Outcomes for Higher Risk Patients Towards a Precision Medicine	Grant (240,000€)	DSAIPA/DS/0117/2020	Researcher

Chapter 6 – Innovation and Technology Transfer

The candidate is the primary author of the provisional patent code **P2134.1 PP**, with the tentative title of “Computer-Implemented Method for an In Vitro or Ex Vivo Diagnosis or Prognosis of a Sample from a Human Patient and Data Processing Device Thereof”, representative of the work developed on biomarkers in the course of the Doctoral Programme and as the evolution of the work started on the Master’s degree. There was also informal support in the establishment of the international patent **PCT/PT2025/050006** “Allograft Rejection Prognosis”. This is complemented by extensive knowledge of industrial property law and inner workings of nationally regulating bodies like INPI.

Chapter 7 – Student Supervision, Teaching Activities & Scientific Outreach

In this chapter it's briefly pointed out the various activities developed while supervising students, from undergraduate to graduate level (Bachelor, Masters and Ph.D.). This list can also be found at <https://rubenaraujo.org/cv/> under "Resume": "Research Focused" and "Activities". They include pro-bono teaching, mentoring and management of IEEE related activities. Other activities like class representative and secretary responsible for all engineering disciplines (e.g. Mechanical Engineering, Civil Engineering, Electronics, etc.) student bodies in ISEL are not represented.

Student Supervision and Mentoring

Over the years, the candidate has provided close support and orientation to 9 Master theses from both ISEL at *Laboratório de Engenharia e Saúde* and IST, as part of collaboration protocols between the two universities. Recently, the supervision has extended to 7 Ph.D. students (5 from the Biomedicine doctoral programme in NOVA Medical School, 1 from IST and 1 from NOVA *Faculdade de Ciências e Tecnologia* (NOVA FCT). Added to these responsibilities, the candidate has supervised and taught, for the last 8 years, laboratory assistants as well as freshmen from the Bachelor in Biomedical Engineering and Chemical Engineering (mostly from ISEL, with the occasional students from IST and Erasmus exchange students from Cape Verde and Latvia (Riga Technical University) .

Teaching Activities

Taught laboratory classes in ISEL for both Bachelor and Master level students, in varied subjects from metabolomics, standard wet lab work, sterilization best practices and use of white rooms, and cell culture. Other work included diverse teaching opportunities such as collaborative events with Lego Group (Ultimate Challenge Week), promoting varied subjects from Robotics and Programming applied to children, as well as 12th grade classes, per invite, at classes in Oeiras International School.

Scientific Outreach

Here it is included diverse activities from Institute of Electrical and Electronics Engineers (IEEE) activities, ISEL and NMS related events. For an extensive list and URLs/PDFs please consult the link mentioned in the beginning of this chapter. Some of these activities include:

- Revival of IEEE student chapter, acting as Chairman (ISEL);
- Creation of EMBS (Engineering, Medicine and Biology Society) chapter in ISEL;
- Welcoming events for Bachelor level classes in Biomedical Engineering;
- Organization of IEEE Leadership camps across the country;

- Participation in the organization body of Biomedical and Health Sciences conferences;
- Regular appearance in interviews on the importance and dissemination of biomedical research;
- Annual conference organization of biomedical engineering students in Portugal;
- Organization of webinars;
- Liaison between student bodies and IEFP/EURES networks;
- Continuous support to this date for Bachelor and Master students in the elaboration of CVs, preparing for professional interviews and in starting their work life;
- Bridge between academia and industry in order to help obtain internships or support for Master Thesis;
- Session moderator in international conferences;
- Speaker in various national and international research conferences;
- Speaker in Big Data and AI in healthcare for international university students.

Chapter 8 – Awards & Distinctions

In this chapter it is presented distinctions and awards obtained during the Master in Biomedical Engineering and the Doctoral Programme in Biomedicine. This list can be found at <https://rubenaraujo.org/cv/> under “Resume”: “Education” and under “Prizes and Awards”, as clickable URLs that lead to downloadable files. These are as follows:

- **First in Class Certificate** (Ph.D. in Biomedicine), with a 19/20 evaluation in the curricular part of the degree (degree was granted by unanimity);
- **First in Class Diploma** (MSc in Biomedical Eng.), recognizing the highest final course average during the two consecutive years of the degree;
- **Merit Scholarship for Excellence and Academic Achievements**, awarded by *Direção Geral do Ensino Superior* (DGES), for the academic year 2018/2019, for an average of 18.38 out of 20, with the award covering 100% of tuition fees.